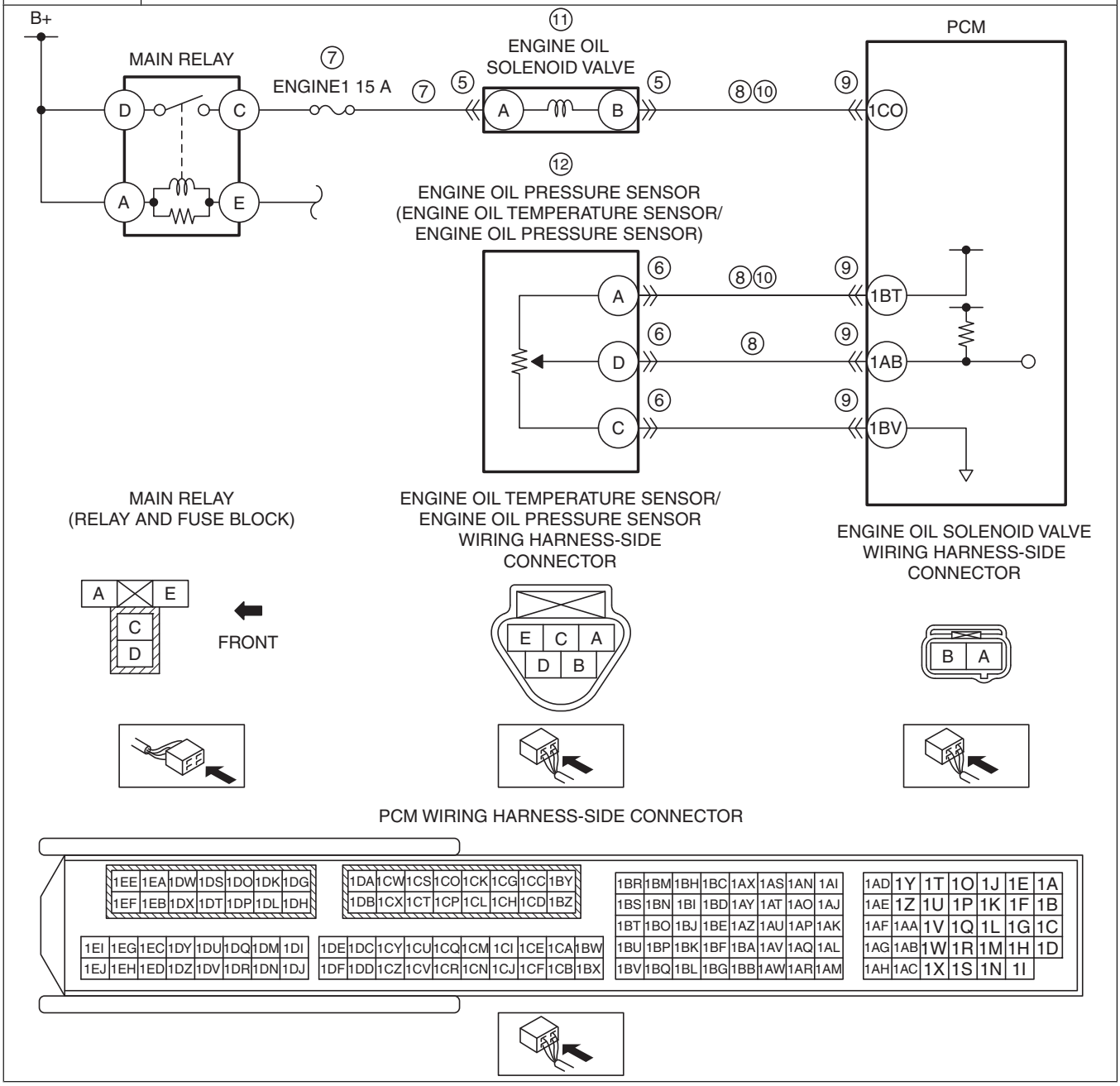
DTC P06DE:00 [SKYACTIV-D 1.5]

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DTC P06DE: 00	Engine oil pressure switch control circuit low input
DETECTION CONDITION	• When the following condition is met, the engine oil pressure is **200 kPa {2.04 kgf/cm^2, 29.0 psi} or less** for **5 s**. **MONITORING CONDITIONS** — Engine oil temperature sensor: Normal — Engine oil pressure sensor: Normal — During high hydraulic pressure control (during engine oil solenoid valve operation) — Engine oil temperature: **20 °C {68 °F} or more** — Engine speed is specified value or more. • Idle speed or more when engine oil temperature is **20 °C {68 °F}** (when cold) • **2,000 rpm or more** when engine oil temperature is **90 °C {194 °F}** (when hot) • **4,000 rpm or more** when engine oil temperature is **135 °C {275 °F} or more** (when hot) — The following DTCs are not detected: • Malfunction due to no engine oil pressure: P0524:00 • Malfunction due to engine oil pressure decrease: P055F:00 **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Engine oil leakage • Improper engine oil level • Engine oil solenoid valve connector or terminals malfunction • Engine oil temperature sensor/engine oil pressure sensor connector or terminals malfunction • Short to ground or open circuit in engine oil solenoid valve power supply circuit — Short to ground in wiring harness between ENGINE1 15 A fuse and engine oil solenoid valve terminal A — ENGINE1 15 A fuse malfunction — Open circuit in wiring harness between main relay terminal C and engine oil solenoid valve terminal A • Short to ground in wiring harness between the following terminals: — Engine oil solenoid valve terminal B—PCM terminal 1CO — Engine oil temperature sensor/engine oil pressure sensor terminal A—PCM terminal 1BT — Engine oil temperature sensor/engine oil pressure sensor terminal D—PCM terminal 1AB • PCM connector or terminals malfunction • Open circuit in wiring harness between the following terminals: — Engine oil solenoid valve terminal B—PCM terminal 1CO — Engine oil temperature sensor/engine oil pressure sensor terminal A—PCM terminal 1BT • Engine oil solenoid valve malfunction • Engine oil pressure sensor malfunction • Oil strainer clogging • Oil pump malfunction • PCM malfunction

DTC P06DE: 00	Engine oil pressure switch control circuit low input



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**INSPECT ENGINE OIL LEAKAGE** • Start the engine. • Verify that there is no engine oil leakage in the hydraulic circuit. • Is there any leakage?	Yes	Repair or replace the malfunctioning part according to the inspection results, then add genuine engine oil. Go to the step 14.
		No	Go to the next step.
4	**INSPECT ENGINE OIL LEVEL** • Inspect the engine oil level. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].) • Is the engine oil level sufficient?	Yes	Go to the next step.
		No	Add genuine engine oil, then go to the next step.
5	**INSPECT ENGINE OIL SOLENOID VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the engine oil solenoid valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.
6	**INSPECT ENGINE OIL TEMPERATURE SENSOR/ENGINE OIL PRESSURE SENSOR CONNECTOR CONDITION** • Disconnect the engine oil temperature sensor/ engine oil pressure sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.
7	**INSPECT ENGINE OIL SOLENOID VALVE POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the engine oil solenoid valve and engine oil temperature sensor/engine oil pressure sensor connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the engine oil solenoid valve terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the ENGINE1 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE1 15 A fuse and engine oil solenoid valve terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and engine oil solenoid valve terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 14.

STEP	INSPECTION		ACTION
8	**INSPECT ENGINE OIL SOLENOID VALVE CIRCUIT AND ENGINE OIL PRESSURE SENSOR CIRCUIT FOR SHORT TO GROUND** • Verify that the engine oil solenoid valve and engine oil temperature sensor/engine oil pressure sensor connectors are disconnected. • Switch the ignition off. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Engine oil solenoid valve terminal B — Engine oil temperature sensor/engine oil pressure sensor terminal A — Engine oil temperature sensor/engine oil pressure sensor terminal D • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — Engine oil solenoid valve terminal B—PCM terminal 1CO — Engine oil temperature sensor/engine oil pressure sensor terminal A—PCM terminal 1BT — Engine oil temperature sensor/engine oil pressure sensor terminal D—PCM terminal 1AB **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 14.
		No	Go to the next step.
9	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.
10	**INSPECT ENGINE OIL SOLENOID VALVE CIRCUIT AND ENGINE OIL PRESSURE SENSOR CIRCUIT FOR OPEN CIRCUIT** • Verify that the engine oil solenoid valve and engine oil temperature sensor/engine oil pressure sensor and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Engine oil solenoid valve terminal B—PCM terminal 1CO — Engine oil temperature sensor/engine oil pressure sensor terminal A—PCM terminal 1BT • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Engine oil solenoid valve terminal B—PCM terminal 1CO • Engine oil temperature sensor/engine oil pressure sensor terminal A—PCM terminal 1BT **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 14.
11	**INSPECT ENGINE OIL SOLENOID VALVE** • Inspect the engine oil solenoid valve. (See ENGINE OIL SOLENOID VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the engine oil solenoid valve, then go to Step 14. (See ENGINE OIL SOLENOID VALVE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
12	**INSPECT ENGINE OIL PRESSURE SENSOR** • Reconnect all disconnected connectors. • Inspect the engine oil pressure sensor. (See ENGINE OIL PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the engine oil temperature sensor/engine oil pressure sensor, then go to Step 14. (See ENGINE OIL TEMPERATURE SENSOR/ENGINE OIL PRESSURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
13	**INSPECT OIL STRAINER** • Inspect the oil strainer for clogging. (See OIL PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the oil strainer, then go to the next step. (See OIL PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Replace the oil pump, then go to the next step. (See OIL PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
14	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Start the engine and warm it up completely. **Caution** **• While performing this step, always operate the vehicle in a safe and lawful manner.** **• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.** • Drive the vehicle under the FREEZE FRAME DATA/snapshot data condition. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
15	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.